

CONCORD, NH, USA

WMO: 726050

Lat: 43.205N

Lon: 71.503W

Elev: 343

StdP: 14.51

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14745

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	1.9	-14.5	2.5	0.6	-10.0	3.2	5.8	25.5	17.7	23.3	16.9	5.7	310	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	90.2	71.6	87.2	69.9	84.3	68.6	74.7	84.8	73.0	82.3	71.4	79.7	9.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	118.7	78.5	70.2	112.6	77.0	68.7	106.7	75.8	38.3	84.6	36.7	82.4	35.4	79.8	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.0	18.6	16.5	-11.2	94.9	6.5	3.0	-15.8	97.1	-19.6	98.9	-23.3	100.6	-28.0	102.8	
			-11.7	77.4	6.3	1.9	-16.2	78.8	-19.9	79.9	-23.4	80.9	-27.9	82.3	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	22.0	24.8	33.2	45.3	56.3	65.5	70.8	69.2	61.4	49.3	38.5	28.3	
	DBStd	18.81	11.06	10.01	9.49	8.30	7.59	6.84	5.38	5.55	7.33	7.82	8.53	9.10	
	HDD50	3458	870	706	528	188	22	0	0	0	3	110	358	673	
	HDD65	7010	1334	1125	986	592	288	77	11	23	154	488	795	1137	
	CDD50	2427	1	0	7	48	218	465	644	595	346	89	13	1	
	CDD65	503	0	0	0	2	19	91	190	152	47	2	0	0	
	CDH74	5436	0	0	14	75	378	1039	1914	1501	474	41	0	0	
	CDH80	1792	0	0	3	22	119	351	679	483	132	3	0	0	
Wind	WSAvg	5.8	6.6	7.1	7.3	7.0	6.1	5.3	4.8	4.3	4.4	5.2	5.6	6.0	
Precipitation	PrecAvg	39.1	2.6	2.6	3.0	3.1	3.3	3.5	3.5	3.5	3.3	3.7	3.5	3.3	
	PrecMax	58.1	8.1	9.0	7.1	7.2	11.1	8.9	7.6	10.7	9.4	14.6	7.5	7.5	
	PrecMin	24.2	0.4	0.0	0.9	0.8	0.5	0.6	1.1	0.0	0.4	0.9	0.5	0.8	
	PrecStd	7.4	1.4	1.5	1.3	1.4	1.9	1.9	1.7	1.9	1.9	2.4	1.5	1.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.4	72.8	83.0	89.3	92.6	93.6	93.1	89.3	79.1	67.5	58.9	
		MCWB	53.7	48.2	56.4	61.1	66.8	73.1	74.4	73.3	71.6	63.6	58.3	51.7	
	2%	DB	47.0	48.7	59.9	73.9	83.3	88.1	90.0	88.4	83.8	73.1	62.3	50.7	
		MCWB	42.2	41.2	48.2	56.0	63.7	70.0	72.3	71.1	68.9	61.4	55.2	46.0	
	5%	DB	40.9	43.1	52.7	67.2	77.8	84.0	86.9	85.3	79.4	67.9	57.6	45.7	
		MCWB	36.6	37.2	43.1	53.1	60.6	67.8	70.7	69.7	66.2	59.3	51.2	41.4	
	10%	DB	36.7	38.8	47.3	61.6	72.7	80.1	83.8	82.2	75.5	63.3	52.7	41.3	
		MCWB	33.1	34.1	40.0	48.7	58.0	65.3	69.1	68.1	64.3	55.9	46.9	37.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	50.5	58.0	63.9	69.4	75.7	77.2	76.3	74.2	68.2	62.0	53.7
			MCDB	54.9	55.2	70.1	75.4	84.4	87.4	89.1	87.2	83.6	73.1	65.8	58.0
2%		WB	42.6	41.8	50.6	59.3	66.5	73.2	74.9	74.0	71.4	64.0	57.2	47.0	
		MCDB	45.8	47.0	57.8	70.1	77.7	83.6	85.1	83.1	79.4	69.7	61.0	49.7	
5%		WB	37.2	37.8	44.4	54.8	63.7	71.0	73.2	72.3	69.3	60.3	52.1	41.9	
		MCDB	40.3	42.4	50.7	64.0	74.0	79.7	82.5	80.6	75.4	66.6	56.0	45.1	
10%		WB	33.5	34.7	40.6	50.4	60.5	68.5	71.5	70.7	67.0	57.0	47.9	37.5	
		MCDB	35.8	38.1	46.7	60.4	69.5	76.0	79.9	78.4	73.0	61.8	52.0	40.6	
Mean Daily Temperature Range	5% DB	MDBR	17.7	19.2	19.4	23.1	23.9	22.5	22.6	23.3	23.5	22.1	18.8	16.7	
		MCDBR	20.8	23.7	28.8	34.6	33.2	28.1	26.6	26.8	27.9	29.0	25.4	22.2	
		MCWBR	17.5	17.5	18.4	18.5	15.8	12.6	11.1	11.7	14.0	17.8	18.4	17.9	
	5% WB	MCDBR	19.1	21.2	25.8	29.4	28.4	23.3	22.8	21.8	22.2	24.2	21.8	20.8	
		MCWBR	16.9	16.6	17.9	18.3	14.9	12.1	10.6	10.6	12.8	16.6	17.8	18.2	
Clear-Sky Solar Irradiance	taub		0.281	0.293	0.319	0.354	0.390	0.433	0.454	0.422	0.370	0.333	0.308	0.286	
	taud		2.431	2.397	2.438	2.410	2.346	2.247	2.195	2.274	2.400	2.516	2.538	2.509	
	Ebn at Noon		273	289	294	290	280	267	259	265	272	271	262	262	
	Edh at Noon		23	29	32	35	39	43	45	40	33	26	22	20	
All-Sky Solar Radiation	RadAvg		543	817	1155	1448	1601	1729	1816	1646	1335	850	578	433	
	RadStd		36	64	74	160	192	163	119	87	85	83	54	46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	N/A
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+61

Nomenclature: See separate page